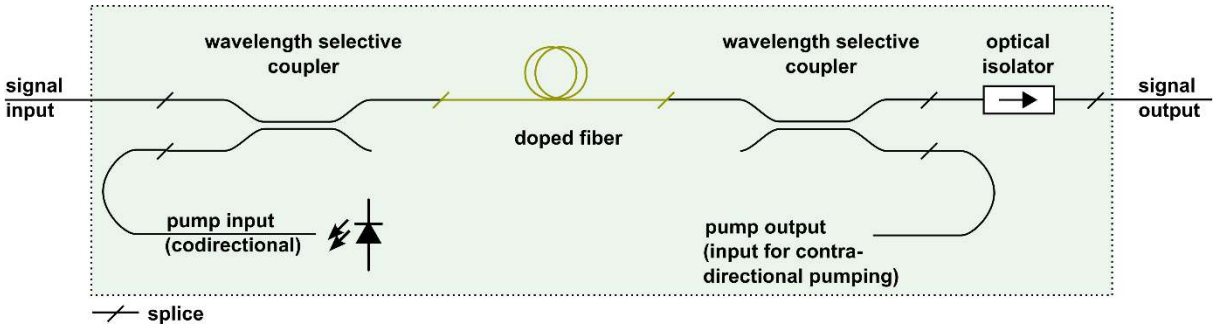
Advanced Optical Fiber Communications

Problem 17: Optical amplifier

For optical amplification at $\lambda = 1.55$ µm, an Er^{3+}-doped fiber with a length of 40 m and a gain coefficient of $g_{pump} = 1.2$ dB/mW is used as shown in the figure below. The pump laser emits 30 mW of optical pump power at $\lambda_{pump} = 980$ nm. Consider fusion splice losses ($\alpha_s = 0.1$ dB per splice), coupling losses for the pump wavelength ($\alpha_p = 0.45$ dB) and the insertion loss of the isolator ($\alpha_{is} = 1$ dB).



a) Determine the net gain G_{fiber} of the fiber amplifier at the emitted pump power.

b) Calculate the total gain G_{total} of the amplifier including splices, couplers and isolator.

c) Calculate the output power P_{out}, when the input signal power is $P_{in} = -30$ dBm.

d) Now, an optical fiber with length L_{oa} is added in front of the input of the optical amplifier. Derive the noise figure for the optical fiber and the optical amplifier, respectively, and calculate the noise figure of the segment for $n_{sp} = 2$. What is the length L_{oa}, if the attenuation of the fiber is 0.2 dB/km and the amplifier compensates the fiber loss?

e) For a long-distance transmission link, several segments need to be used. What is the amount of segments N for an OSNR of 30 dB, $P_0 = 1$ mW, and $L_{oa} = 30$ km?

Problem 18: Submarine transmission link

A new state of the art submarine transmission link between France and USA shall be designed. The total length of the link is 6600 km and the desired total data rate is 200 Tbit/s. The number of wavelength division multiplex channels is 50 and with a novel few modes optical fiber 10 modes are supported.

a) What is the data rate per mode and channel that needs to be achieved to get the desired total data rate?

b) The segment length is L_{oa} = 100 km. Calculate the OSNR in dB with P_0 = 10 mW, F_{amp} = 4, and α_{dB} = 0.2 dB/km.

c) The forward error correction limit is $1 \cdot 10^{-3}$ and the used code has an overhead of 7%. Which modulation scheme shall be used in order to minimize the used channel bandwidth, when the SNR shall not exceed 20 dB? Calculate the symbol rate.

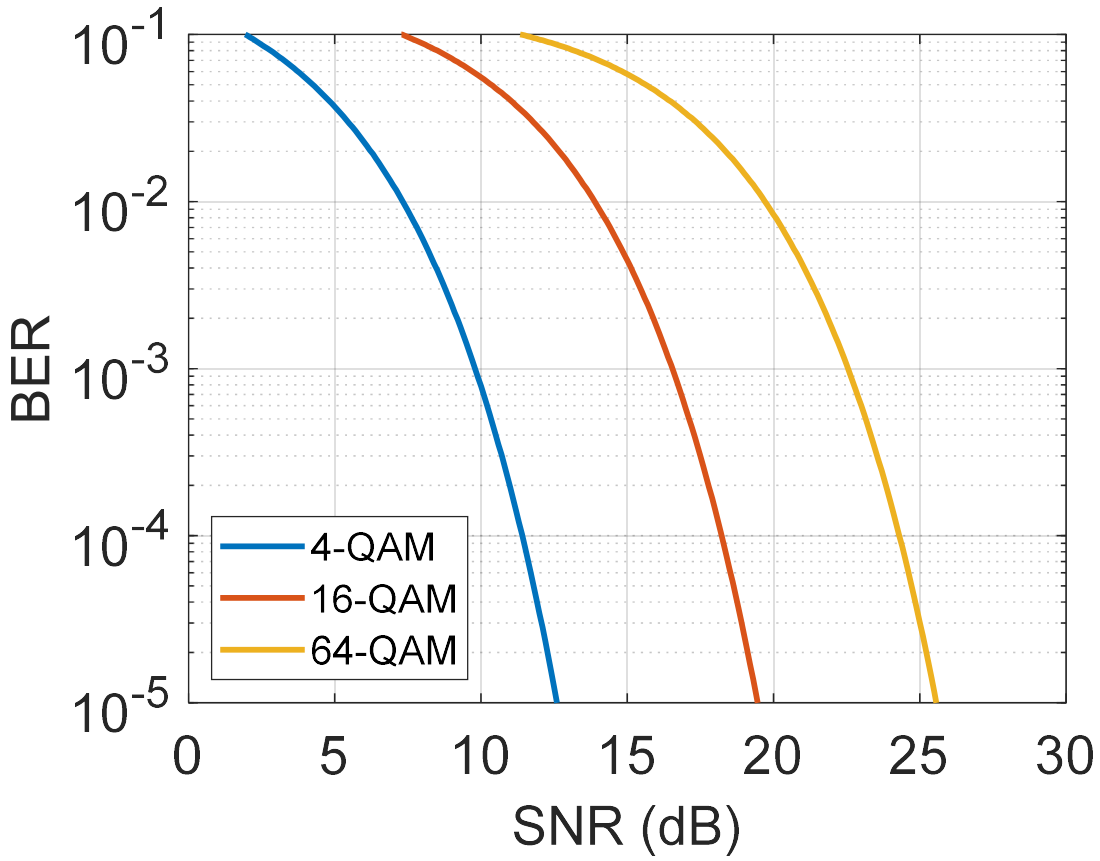
d) Calculate the required OSNR in dB based on the found results in c). Is it possible to use the transmission link for the found modulation scheme?

e) In order to save costs, the number of amplifiers shall be reduced, hence, the segment length is increased to 120 km. Is the required bit rate achievable with this modified transmission link?

A coherent receiver is necessary to demodulate the transmitted QAM signal.

f) Sketch a diagram of the receiver and derive the output current of the in-phase and quadrature component.

g) As seen in part e), the costs can be reduced with a smaller number of segments. How can the costs be lowered even further on the receiver side?


BER
SNR (dB)
4-QAM
16-QAM
64-QAM
10^-1
10^-2
10^-3
10^-4
10^-5
0
5
10
15
20
25
30